

Figure 1: Flux features

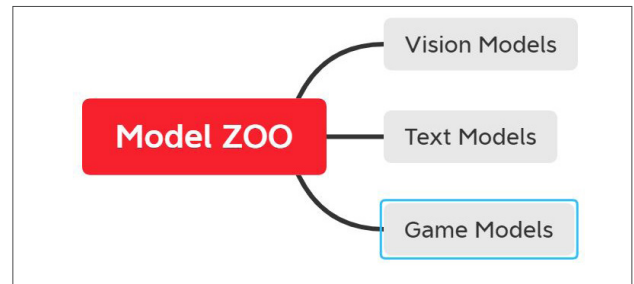


Figure 2: Flux model Zoo

The Flux ecosystem

One of the critical factors that will push you to choose Flux over other tools is the availability of an ecosystem which is very diverse in nature.

- With the help of libraries such as *Turing.jl*, Bayesian inference and Gaussian processes can be handled on top of Flux.
- Geometric deep learning, which is gaining a lot of attention these days, can be performed through GeometricFlux.
- For computer vision tasks, Metalhead shall be utilised. In the case of transfer learning, these models are highly useful.
- The transformers.jl is adapted for transformer models for natural language processing (NLP) tasks.

Getting started with Flux

Make sure that Julia is installed in your system. The Julia version should be at least 1.3. You can then install Flux with the following simple command in the Julia REPL:

```
julia> ] add Flux
```

An alternative method is as shown below:

```
julia> using Pkg; Pkg.add("Flux")
```

The Flux model Zoo

The Flux model Zoo contains a diverse demonstration of Flux. It can be adapted for the objective of your applications as well. The models provided by Flux Model Zoo can be grouped into the following categories:

- Vision models (These are very large convolutional neural networks or CNNs.)
 - Text models (though various recurrent neural networks or RNNs)
 - Games (through reinforcement learning)
- Complete details on the model Zoo can be obtained by navigating to <https://github.com/FluxML/model-zoo>

Steps involved

This section explores the steps involved in building a solution through Flux.

- Provide training and test data
- Construct a model with hyper parameters
- Iterative training of the model to improve performance
- Verification of the model

As stated earlier, a significant advantage of Flux is that it enables you to choose the abstraction level. If you prefer to perform each step yourself, then too Flux makes it simple enough to do so. In this section, we will go through building and training a model using Flux in a step-by-step manner. The model will predict the output of the function $2x + 3$. The

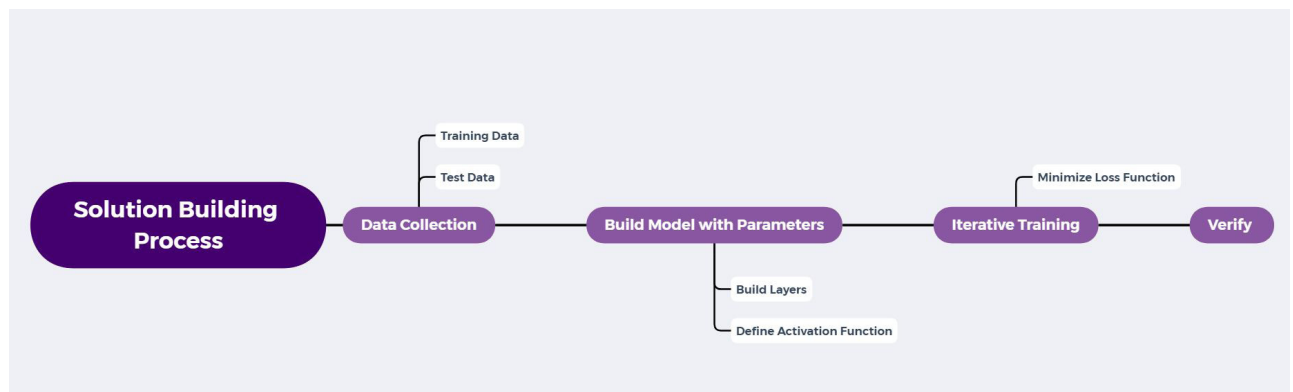


Figure 3: Solution building process